

AKASH SARAVANAN

GEOSPATIAL ENGINEER | GIS · WEB GIS · GEOAI

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PROFILE SUMMARY

Geospatial Engineer with nearly four years of experience delivering GIS mapping, remote sensing, spatial analysis, and geodatabase management across infrastructure and land-development projects. Specialised in QGIS, ArcGIS Pro/Online, Google Earth Engine, and Python-based geospatial automation (GDAL, Rasterio, GeoPandas, PyQGIS). Leverage AI tools (Claude, ChatGPT) and AI-assisted development to build Web GIS platforms end to end and to automate day-to-day GIS workflows, and apply GeoAI and machine-learning models for land-use analysis and automated feature extraction. Led QA/QC processes achieving 100% data integrity across cadastral and property-tax geospatial projects while mentoring a two-person team. Focused on translating complex spatial data into accurate, decision-ready outputs that support infrastructure planning and land-development decisions.

Highlights: 30+ CMDA layout projects on a Web GIS platform I built · 700+ parcels and 5,000+ acres mapped · 25% QC workload cut · 100% data integrity on property-tax projects · 13 of 15 parcels auto-placed within 5 m

TECHNICAL SKILLS

- GIS & Mapping Tools: ArcGIS Pro, ArcGIS Online, Field Maps, QGIS, Google Earth Engine, AutoCAD Map 3D, Google Earth Pro, Global Mapper
- Remote Sensing & Earth Observation: Google Earth Engine, satellite imagery analysis, vegetation indices (NDVI), image classification, GeoAI / machine learning for feature extraction and land-use prediction
- Data & Geodatabase Management: Geodatabase structuring, metadata documentation, folder structure/naming conventions, version control, vector/raster conversion, georeferencing, geoprocessing
- BIM – GIS Integration: Coordinating GIS/Spatial data with CAD & BIM drawings, leveraging civil engineering domain knowledge.
- Development & Automation: Python (GDAL, Rasterio, GeoPandas, Shapely, PyQGIS — geospatial workflow automation), ModelBuilder, FME, PostGIS, SQL; JavaScript (React, Leaflet, MapLibre GL) for Web GIS with AI-assisted development; LLM APIs (Gemini) for image-based verification
- Drone Data Processing: Pix4Dmapper, Agisoft Metashape, RealityCapture, DJI Terra
- Spatial Analysis & Cartography: Multi-criteria spatial analysis, spatial data analysis, cartographic map production, symbology and layout design
- Productivity: Microsoft 365 (Excel, PowerPoint, Word)

EXPERIENCE

FARMWISEAI PRIVATE LIMITED – Geospatial Engineer; Chennai, India

September 2025 – Present

- Techno-Economic Feasibility Studies (TEFR): Conducting spatial and economic analysis for proposed industrial parks — land potential, infrastructure connectivity and investment viability — and coordinating GIS datasets with CAD/BIM survey drawings so layouts stay consistent across disciplines.
- Site Suitability Modeling: Developed multi-criteria decision-making models to identify optimal locations for industrial setups, balancing technical constraints with environmental and logistical factors.
- Precision Drafting & Layout: Executing end-to-end AutoCAD FMB (Field Map Book) stitching, transforming raw survey data into complete, high-accuracy industrial layouts — 700+ parcels stitched and 5,000+ acres mapped to date.
- Quality Assurance (QA/QC): Leading the quality control process for Property Tax Geospatial Projects using QGIS, ensuring 100% data integrity between physical spatial records and tax databases.
- Web GIS Platform Development: Built the CUMTA (Chennai Unified Metropolitan Transport Authority) Web GIS platform end to end using AI-assisted development (Claude, ChatGPT) — a React/Leaflet map viewer, a FastAPI backend with role-based access, and a CAD pipeline that converts DWG/DXF layouts into georeferenced GIS data for 30+ CMDA layout projects.

- **FPO Portal Contribution:** Supporting ongoing development of an FPO (Farmer Producer Organization) platform — contributing to frontend UI improvements and independently building land-document retrieval workflows (Patta/Chitta, A-Register) to feed accurate cadastral data into the farmer database.
- **GIS Automation & GeoAI:** Streamlining daily GIS workflows with Python scripts and tools that cut manual data entry, and applying GeoAI / machine-learning models to spatial data for predictive land-use analysis and automated feature extraction.
- **Stakeholder Presentation & Field Data:** Translating technical suitability reports into visual, data-driven presentations for executive decision-making, and using ArcGIS Online dashboards and Field Maps for site assessment and survey capture.

OPERATIONAL ENERGY GROUP INDIA LIMITED – Chennai, India

January 2023 – August 2025

I. OEG DEFENCE AND AEROSPACE PRIVATE LIMITED – GIS Analyst;

Jun 2024 – Aug 2025

- Developed GIS datasets and topographic maps for engineering applications.
- Produced cartographic outputs (site maps, layout plans, thematic maps) using QGIS for engineering review and client submission.
- Used ArcGIS Online and Field Maps for field data collection and visualization to support project data workflows.
- Ensured positional accuracy and consistency of map outputs through systematic QA/QC checks.

II. OPERATIONAL ENERGY GROUP INDIA LIMITED – GIS Analyst;

Jan 2024 – May 2024

- Conducted volumetric analysis and topographic mapping for thermal power plant sites.
- Prepared land survey-based mapping deliverables to support infrastructure planning.
- Maintained mapping standards and symbology conventions across project deliverables.

III. ZUPPA OEG GEN5 TECHNOLOGIES PRIVATE LIMITED – GIS Analyst;

Jan 2023 – Dec 2023

- Processed drone-captured imagery into georeferenced Orthomosaics and digital elevation models for site mapping.
- Generated topographic outputs from UAV survey data for engineering and client review.
- Supported geospatial data integration between drone-derived datasets and GIS platform (QGIS) for mapping deliverables.
- Provided mapping-related technical support and demonstration at client meetings.

SSG ENGINEERS – Site Engineer; Salem, India

December 2021 – December 2022

- Monitored daily construction activities to ensure adherence to project plans and safety standards.
- Coordinated with contractors and suppliers to maintain efficient workflows and timely material deliveries.
- Maintained accurate documentation of project progress inspections and site reports.

EDUCATION

PANIMALAR ENGINEERING COLLEGE – Chennai, INDIA

September 2020

Bachelor of Engineering, Civil Engineering; Cumulative GPA: 7.32/10.0

THIAGARAJAR POLYTECHNIC COLLEGE – Salem, INDIA

May 2017

Diploma, Civil Engineering; Cumulative Percentage: 86.74

PROJECTS

- **CUMTA Web GIS Platform (PoC):** Built a full-stack Web GIS platform for the Chennai Unified Metropolitan Transport Authority to manage CMDA layout data across 30+ layout projects, using AI-assisted development (Claude, ChatGPT). Delivered the React/Leaflet viewer, a FastAPI backend with JWT role-based access, and a CAD ingestion pipeline (ezdxf, ODA File Converter) that converts DWG/DXF layouts into georeferenced GeoJSON with GCP-based correction and version rollback. Added a road register with connectivity and width-compliance analysis, exports to Shapefile, KML, Excel and PDF, and a dual SQLite/PostGIS data layer, backed by 250+ automated tests.

- **FPO Portal:** Map-first management cockpit for Farmer Producer Organizations, where a survey-level land map and a live MIS dashboard work as one instrument. Integrated real government data (land parcels, soil-health cards, fertilizer dealer stock) and built the village-drill and parcel-selection experiences on MapLibre GL with statewide PMTiles layers for a 323-member FPO in Tamil Nadu.
- **Automated FMB Georeferencing, Tambaram–Chengalpattu Rail Corridor (R&D, open source):** Building a Python pipeline that converts Tamil Nadu Field Measurement Book (FMB) PDFs into vector parcels with every printed dimension preserved, then places each sheet on the ground by matching shared boundaries with already-placed neighbours (rotation and shift only, scale fixed at 1), with green/amber/red confidence flags for review. In leave-one-out validation on a calibration village, 13 of 15 parcels were re-placed within 5 m. Built with GeoPandas, Shapely, Rasterio, OpenCV and SciPy, with 120+ unit tests.
- **Property-Tax Street Audit, Coimbatore City Municipal Corporation (Proof of Concept, open source):** Built a Python tool that checks property-tax building records against street-level imagery: it picks panoramas with a real line of sight to each footprint, marks the target, and has a vision LLM (Gemini) read floors, usage and shop fronts blind, while fixed rules decide the verdict and compare it with the survey and tax record.
- **PyQGIS QC Automation, Coimbatore City Municipal Corporation:** Built a PyQGIS tool that flags spatial mismatches between building-footprint and geocode shapefiles across the citywide property-tax dataset (residential, commercial and industrial), with custom in-QGIS shortcuts that take reviewers straight to each flagged feature. Reduced manual quality-control workload by 25%.
- **Site Suitability & Techno-Economic Feasibility (TEFR) for Proposed Industrial Parks:** Conducted multi-criteria spatial analysis integrating vector (land use, infrastructure proximity) and raster (terrain, environmental) datasets into suitability maps for data-driven site selection, and contributed the geospatial inputs — GIS-based layouts and location intelligence — to techno-economic feasibility reports that informed planning, investment viability and regulatory compliance decisions.
- **Cadastral FMB Stitching & Large-Scale Land Parcel Mapping:** Executed AutoCAD-based FMB stitching workflows across multiple project sites — 700+ parcels stitched and 5,000+ acres mapped, on sites ranging from 200 to 2,000 acres. Ensured spatial accuracy and boundary integrity for cadastral records, supporting land acquisition and legal documentation processes in large-scale infrastructure projects.
- **Photogrammetry Mapping & Volumetric Estimation (Highways, Railways & River Beds):** Processed aerial survey data to generate DSMs, orthomosaics, and contour maps for National Highways, railway corridors, and river-bed projects. Performed volumetric estimation for coal stockpiles using point cloud and elevation data, delivering high-accuracy outputs for infrastructure planning and resource quantification.

KEY COMPETENCIES

- GIS · Remote Sensing · Spatial Analysis · Photogrammetry · Cadastral Mapping · Geodatabase Management
- Web GIS · GeoAI / Machine Learning · Python · JavaScript · PostGIS · Data Analysis
- Data QA/QC · GIS Project Planning & Delivery · Stakeholder Communication · Cross-functional Collaboration

CERTIFICATIONS AND TRAINING

- AutoCAD Certification (2017)
- AI/ML for Geodata Analysis — Indian Institute of Remote Sensing (IIRS), ISRO (2024)
- Essentials for PyQGIS: Python for Geospatial Automation — Udemy (2025)
- Drone Photogrammetric Image Processing — Udemy (2024)
- GIS Basics — Esri (2023)
- ArcGIS Pro Essential Training — LinkedIn Learning (2023)